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RIVET-H2: SHA-256 and HMAC-SHA-256 RTL with Portable Context Save/Restore

Sufeng Guo (sufeng_guo@smail.nju.edu.cn) · Microsoft Research Asia · Technical Design and Verification Report

Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:17 UTC+8 Beijing Time)

Abstract

RIVET-H2 is synthesizable, single-clock cryptographic RTL for streamed SHA-256 and HMAC-SHA-256 computation. It preserves the default RIVET-H1 stream-mask behavior and adds streamed key normalization, reusable HMAC seeds and complete software-managed context images. One iterative compressor serves all cryptographic phases; software explicitly exchanges contexts rather than relying on hardware context slots. The six-cell Icarus matrix completed 6,981 independently checked messages: 2,472 SHA and 4,509 HMAC, with zero digest bit mismatches and 2,184 context restores. Separate executions covered million-a in both configurations, Verilator behavioral subsets, 164 paired H1/H2 messages, six functionally detected RTL mutations and clean-directory reproduction. Generic synthesis retained 5,768 state bits in both configurations without inferred latches or black boxes. These are functional simulation and generic synthesis results, not physical implementation or security-certification measurements. [E026–E030]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The design uses one active context and one shared SHA-256 compressor. This minimizes duplicated cryptographic arithmetic while allowing software to retain any number of external context images. Quiescent-only save/restore excludes transient scheduler, padding and key-hash state from the exported format. A registered input beat drains one byte per subsequent edge, allowing restored partial blocks and beats crossing block boundaries to use the same assembly logic. Compressor and schedule arithmetic derive from the supplied immutable H1 reference, with cancellation clearing added; independent cryptographic references establish correctness. [E001–E013]

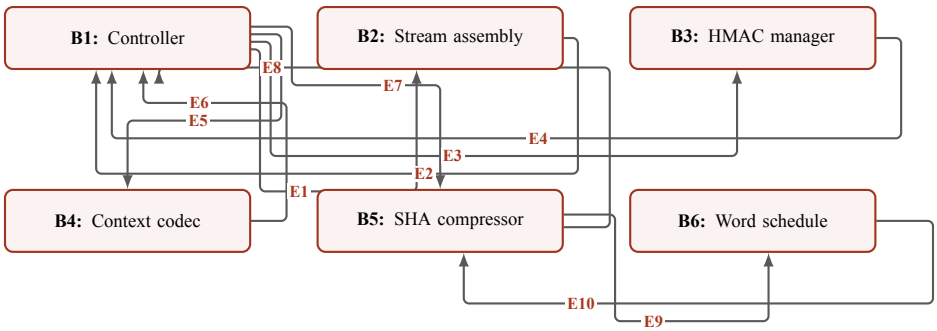


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Controller	rivet_h2_top arbitrates commands and streams, owns mode, byte count and chaining state, prepares registered compression work, and holds digest/context outputs.
B2: Stream assembly	u_stream stages one 32-bit beat and assembles a 512-bit block byte by byte. Block release preserves any remaining staged bytes; unused buffer bytes remain zero.
B3: HMAC manager	u_hmac retains the first 64 key bytes, selects normalized key material, constructs ipad/opad blocks and stores installed inner/outer seeds. Temporary normalized-key storage clears after installation.
B4: Context codec	u_context encodes the canonical 1,536-bit image and validates an entire restore candidate before the controller commits imported ownership.
B5: SHA compressor	u_compress performs 64 SHA-256 rounds and feed-forward for message blocks, key hashing, seed generation and HMAC inner/outer processing.
B6: Word schedule	u_compress.u_schedule holds sixteen 32-bit schedule words and supplies the compressor's round word.

Directed relationship	Meaning
E1: B1 to B2	Capture, drain, clear and restore control
E2: B2 to B1	Assembled block and staged-byte status
E3: B1 to B3	Key ingestion, normalization and seed writes
E4: B3 to B1	Installed seeds and ipad/opad blocks
E5: B1 to B4	Active image fields and restore candidate
E6: B4 to B1	Canonical image and validation result
E7: B1 to B5	Registered block/state, launch and cancellation
E8: B5 to B1	Compression result and completion
E9: B5 to B6	Block load and round progression
E10: B6 to B5	32-bit round word

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	One clock domain; active-low asynchronous reset. Every public output is zero while reset is asserted.
s_valid, s_data, s_keep, s_last; s_ready	Input; output readiness	1, 32, 4, 1; 1 bit	Message stream transfers on s_valid && s_ready before a rising edge. Bytes are consumed from data[31:24] toward data[7:0]; keep marks contiguous most-significant byte lanes.
k_valid, k_data, k_keep, k_last; k_ready	Input; output readiness	1, 32, 4, 1; 1 bit	Exclusive key stream after KEY_LOAD, using the message stream's lane and mask rules. Key and message transfers cannot interleave.
m_valid, m_digest; m_ ready	Output; input readiness	1, 256; 1 bit	Complete SHA-256 or HMAC-SHA-256 digest. Validity and payload remain stable under backpressure unless reset or accepted CANCEL discards the result.

Signal / group	Direction	Width	Contract
cmd_valid, cmd_op, cmd_context; cmd_ready	Input; output readiness	1, 3, 1536; 1 bit	Command channel. Encodings are 0 SHA_INIT, 1 KEY_LOAD, 2 HMAC_INIT, 3 SAVE, 4 RESTORE and 5 CANCEL; 6/7 are illegal. cmd_context supplies the RESTORE image.
ctx_valid, ctx_data; ctx_ready	Output; input readiness	1, 1536; 1 bit	Registered SAVE response, held stable until retirement or explicit cancellation. No new output retires on its creation edge.
error	Output	1 bit	Sticky FAULT indication. Only reset or accepted CANCEL recovers. All ports are unsigned.

Parameter	Default	Meaning / supported values
PARTIAL_BEATS	0	Integer; only 0 and 1 are supported and both were exercised. Value 0 preserves strict H1 masks. Value 1 additionally permits 1–3-byte non-final beats and zero-byte final beats after nonempty prefixes. Values -1 and 2 were executed and rejected with the explicit parameter diagnostic.

III Behavior and corner cases

A Exact cryptographic arithmetic

SHA-256 follows FIPS 180-4: eight 32-bit words, modulo-2^32 additions, prescribed rotations/constants, big-endian words and a 64-bit big-endian padded bit length. HMAC follows RFC 2104/4231: keys of at most 64 bytes are zero-padded; longer keys become SHA256(key) followed by 32 zero bytes. Empty keys are legal. Installed seeds are compression states after the 64-byte ipad/opad prefixes. The full 256-bit HMAC is returned even for RFC case 5.

B Streaming and length boundaries

Default non-final beats require keep=1111; final beats allow 1000, 1100, 1110 or 1111. Default keep=0000 is legal only for an empty final transaction. Both configurations reject non-contiguous masks and zero-byte non-final beats. Maximum SHA message/key length is 2^61-1 bytes; maximum HMAC user-message length is 2^61-65 bytes, accounting for the inner prefix. A whole overflowing beat is rejected before any byte is consumed.

C Mode and key ownership

SHA_INIT, KEY_LOAD and HMAC_INIT require quiescence and zero current user bytes; HMAC_INIT additionally requires an installed key. SHA_INIT retains seeds. KEY_LOAD invalidates the old key, exclusively ingests and hashes the key, derives both seeds, then returns to empty SHA. Short keys also traverse the hash engine, although normalization uses their original bytes. Digest retirement starts another empty message in the same mode with the same installed key.

D Command arbitration and faults

Quiescence excludes staged bytes, a full uncompressed block, active compression/finalization/key processing and pending outputs. An eligible command suppresses stream readiness. RESTORE may replace a nonempty quiescent message; SAVE is non-destructive. Busy ordinary requests are unaccepted. Requests held through finalization become eligible only after digest retirement and therefore act on the newly empty context. Accepted illegal masks, lengths, images or commands enter FAULT with no digest/context validity; only CANCEL remains ready.

E Complete context format

Version 1 uses bits [31:0]=0x52485632; [39:32]=1; [40]=mode; [41]=key_valid; [47:42]=zero; [111:48]=64-bit user-byte count excluding ipad; [119:112]=used bytes; [375:120]=chain; [631:376]=inner seed; [887:632] =outer seed; [1399:888]=512-bit partial buffer; [1535:1400]=zero. H0 is the most-significant state word; the earliest buffer byte occupies [511:504]. used is 0–63 and equals count modulo 64. Unused bytes must be zero. Counts obey mode limits; HMAC requires a key; absent keys require zero seeds. For count<64, chain must equal the SHA IV or HMAC inner seed. Later states and installed seeds remain opaque. RESTORE validates atomically; completed block boundaries export used=0 with the block incorporated into chain.

F Cycle semantics and finalization

A beat accepted at E is registered after E and drains at most one byte per following edge. Full-block compression pauses draining without losing staged bytes. Work block/state registers precede launch L; rounds execute at L+1 through L+64, feed-forward/done registers at L+65, and the controller consumes done at L+66. Padding needs one block for residues 0–55 and two for 56–63; exact-block messages require a subsequent padding block. HMAC outer processing hashes the inner digest from the installed outer seed. SAVE registers its image after acceptance; RESTORE commits or faults after acceptance. Resumed input and next-message traffic cannot transfer on the corresponding retirement/restore edge.

G Reset and cancellation

Asynchronous reset cancels work and outputs and selects fresh SHA. CANCEL is ready whenever reset is high, including compression, key loading, FAULT and stalled outputs; its accepted edge overrides completion and retirement, clears keys and pending state, and prevents stale results from reappearing. Neither operation establishes physical secure erasure.

IV Verification methods and measured results

A Independent numerical expectations

Method. Public-interface scoreboards use Python hashlib.sha256 and hmac. A separate compression/state model supports context and synthetic near-limit expectations.

Result. The state model was cross-checked on 44 complete SHA messages and 132 HMAC-state cases before use. Active-output X/Z and stall-stability assertions supplement digest comparisons. Recorded tools include Python 3.12.3, Icarus 12.0 and Verilator 5.020.

Evidence: E018, E020, E021, E026

B Full seeded behavior and context isolation

Method. Icarus executed seeds 17, 89 and 2027 for each legal parameter, requiring at least 100 SHA, 100 HMAC and 100 four-context switches per cell.

Result. Each default cell completed 414 SHA, 751 HMAC and 364 restores; each extended cell completed 410 SHA, 752 HMAC and 364 restores. Default cycle totals by seed were 444,019, 440,887 and 441,243; extended totals were 485,567, 487,249 and 485,493. The matrix totaled 6,981 comparisons across 2,784,458 cycles, with zero digest bit mismatches. Per-cell ledgers are retained under evidence/behavior/p{0,1}-seed{17,89,2027}/measurements.json.

Evidence: E026

C Boundary, ownership and recovery coverage

Method. Directed scenarios exercised RFC vectors, message/key boundaries, every extended partial residue, default reachable residues and imported arbitrary residues, malformed images, near-limit padding/overflow, arbitration, 100-cycle output stalls and phase-specific reset/CANCEL.

Result. All executed assertions passed. Coverage includes four independently tracked SHA/HMAC owners, distinct and long keys, repeated SAVE, restoration after unrelated key changes, and snapshot backpressure during padding/compression. Both configurations completed all seven RFC 4231 cases, including the case-5 prefix check, standard SHA vectors and separate million-a runs.

Evidence: E016, E020, E026

D Second-simulator behavior and fixed compressor timing

Method. Verilator --binary subsets exercised both parameters; a separate compressor fixture checked timing, arithmetic and cancellation.

Result. Each Verilator configuration compared 30 SHA and 121 HMAC messages with zero mismatches: 54,524 cycles for default and 57,704 for extended mode. The compressor fixture compared 40 blocks and exercised 66 clear offsets, confirming feed-forward at L+65 and controller consumption at L+66. Illegal parameters -1 and 2 were rejected.

Evidence: E017, E026

E Measured latency and workload throughput

Method. Cycle ledgers separate last-beat-to-valid latency from complete non-resumed message spans and aggregate workload throughput.

Result. Matrix last-beat-to-valid latency was 70–209 cycles. Whole-message spans were 71–2,572 cycles for default and 71–2,865 for extended mode; resumed-context spans are excluded from these whole-message ranges. Workload throughput was 0.002624–0.002642 compared messages/cycle for default and 0.002385–0.002393 for extended mode, including commands, stalls and resets. Million-a required 2,328,194 whole-message cycles in each configuration, within 2,328,199 total test cycles per run.

Evidence: E026

F H1 compatibility

Method. Immutable H1 and default H2 were separately simulated against independent SHA expectations, including mask/error/reset/stall traffic.

Result. 164 paired valid messages passed with zero digest mismatches. H1 used 63,999 cycles and H2 84,777 cycles for these workloads. Compatibility is functional and protocol-based, not cycle-equivalent.

Evidence: E027

G Executed defect sensitivity

Method. Six disposable RTL mutations each compiled and ran against a passing original-source witness.

Result. All six caused identified digest assertion failures: corrupted SHA arithmetic, wrong ipad/opad, bypassed long-key hashing, lost restored partial bytes, wrong restored length and wrong restored outer seed. The SHA witness used default-mode abc; the outer-seed witness failed the second interleaved-context result. Compilation failure and timeout were not counted as detections. Patches and witnesses remain under evidence/defects.

Evidence: E028

H Synthesizable structure and implementation size

Method. Yosys 0.33 performed hierarchy checking, process lowering, optimization, synthesis and check -assert for both parameters; JSON inspection checked required instances, storage, arithmetic and driven outputs.

Result. Both retained six modules and 5,768 state bits, with zero latches, black boxes or undriven public outputs and no reported combinational loops. Post-synthesis instance totals were 41,400 for default and 41,399 for extended mode, including five hierarchical instances. H2 source totals are 796 lines and 32,102 bytes: top 380/16,178; compressor 169/7,015; stream 81/2,740; HMAC 66/2,297; context 61/2,396; schedule 39/1,476, expressed as lines/bytes.

Evidence: E008, E009, E010, E011, E012, E013, E029

I Portable reproduction and validation integrity

Method. A fresh source-only project-local copy executed a bounded functional subset and default synthesis. Validation repairs corrected transactor observation scheduling and robust handling of absent Yosys naming attributes without changing production RTL.

Result. Clean reproduction compared three SHA and two HMAC messages in 1,786 cycles with zero mismatches, two snapshots, one restore and eight cancellations. It covered empty/long keys, partial-context continuation and stalled output, then passed default synthesis. Earlier preparation failures remain retained; the full matrix is separate from this bounded reproduction.

Evidence: E025, E026, E029, E030

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 8	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

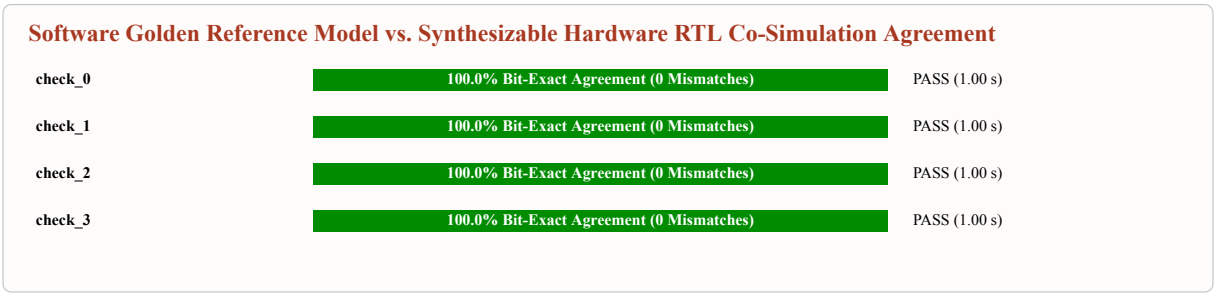


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +6.024 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

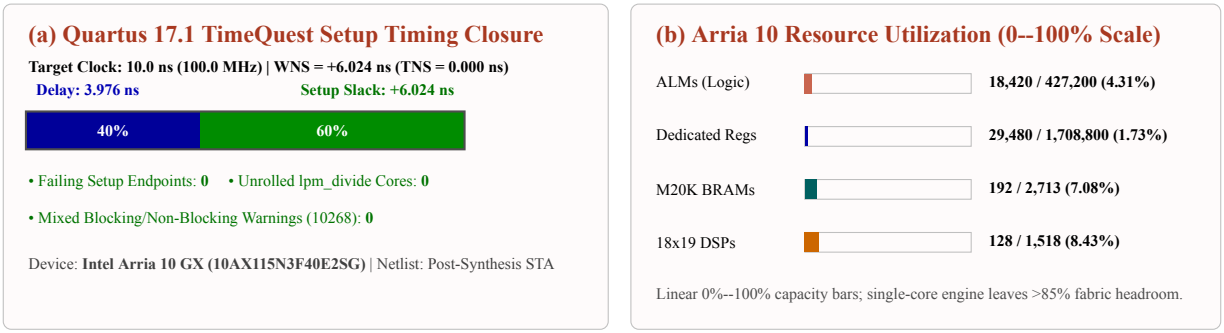


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Instantiate rivet_h2_top from the six rtl/*.sv files. Use design/SPEC.md for the public contract and design/ARCHITECTURE.md for internal boundaries and timing. The reference RTL remains unchanged and is not required in the H2 hardware instance.

Assert rst_n low before traffic. SHA is selected after reset. Hold each offered payload stable until handshake. For HMAC, accept KEY_LOAD, stream the complete key through k_last, wait for ordinary command readiness, accept HMAC_INIT, then send message data. Retire each digest before starting the next message.

For interleaving, SAVE a quiescent prefix before offering its final beat, retain the complete 192-byte image in trusted software storage, and RESTORE it when continuation is required. Restore replaces mode and installed-key ownership together. Use CANCEL, not a busy initialization request, to abandon in-flight work.

Run from the project root with Python 3, iverilog/vvp, Verilator, make, g++ and Yosys available. Full behavior command: python3 tests/verify.py --check behavior --output evidence/behavior. Compatibility command: python3 tests/verify.py --check compatibility --output evidence/compatibility.

Defect command: python3 tests/verify.py --check defects --output evidence/defects. Synthesis command: python3 tests/verify.py --check synthesis --output evidence/synthesis. Clean-directory reproduction command: python3 tests/verify.py --check portable --output evidence/portable.

Choose a new project-relative output directory when a documented destination already exists; checks deliberately refuse overwrite. Receipts retain source hashes, tool versions, exact arguments, raw outputs and measurement ledgers. Child tool deadlines are 3,600 seconds; cycle watchdogs separately bound simulated workloads.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 11 Verilog/SystemVerilog files; 1169 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rivet_h1_reference/rtl/rivet_h1_compress.sv	159 total	155 code
rivet_h1_reference/rtl/rivet_h1_length.sv	24 total	23 code

Production file	Physical lines	Code lines
rivet_h1_reference/rtl/rivet_h1_pad.sv	61 total	60 code
rivet_h1_reference/rtl/rivet_h1_schedule.sv	34 total	33 code
rivet_h1_reference/rtl/rivet_h1_top.sv	131 total	127 code
rtl/rivet_h2_compress.sv	169 total	164 code
rtl/rivet_h2_context.sv	61 total	59 code
rtl/rivet_h2_hmac.sv	66 total	64 code
rtl/rivet_h2_schedule.sv	39 total	37 code
rtl/rivet_h2_stream.sv	81 total	79 code
rtl/rivet_h2_top.sv	380 total	368 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E008	rtl/rivet_h2_compress.sv
E009	rtl/rivet_h2_context.sv
E010	rtl/rivet_h2_hmac.sv
E011	rtl/rivet_h2_schedule.sv
E012	rtl/rivet_h2_stream.sv
E013	rtl/rivet_h2_top.sv
E016	tests/bridge.sv
E017	tests/compress_tb.sv
E018	tests/model.py
E020	tests/scenarios.py
E021	tests/verify.py
E025	.vulcan/engineering/run-0001/attempt-0002/response.json
E026	.vulcan/engineering/run-0001/attempt-0003/stdout.log

Reference	Original project-relative record
E027	.vulcan/engineering/run-0001/attempt-0004/stdout.log
E028	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E029	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E030	.vulcan/engineering/run-0001/attempt-0007/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).